

Business Performance Analysis and Service Optimization: A Primary Data Study of a Professional Laundry Business.

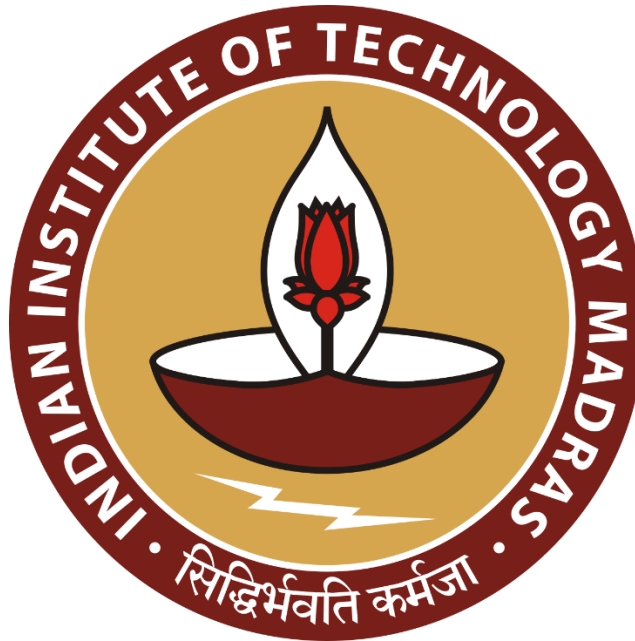
A PROPOSAL REPORT FOR THE BDM CAPSTONE PROJECT

Submitted by:

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(BS) DEGREE IN DATA SCIENCE AND APPLICATIONS

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1. Declaration Statement

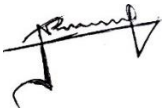
I am working on a project titled "Business Performance Analysis and Service Optimization: A Primary Data Study of a Professional Laundry Business". I extend my sincere gratitude to the management of RDS Associates, MM Towers, Fathima Nagar, East Fort, Thrissur, Kerala for providing the necessary operational data and business information required to conduct this project.

I hereby assert that the data presented and assessed in this project report is genuine and accurate to the best of my knowledge and capabilities. The data has been gathered directly from the business through primary data collection and has been carefully documented to ensure its reliability.

I affirm that all procedures employed for the purpose of data collection and analysis will be duly explained in this report. The outcomes and inferences derived from the data will be an accurate depiction of the findings obtained through the analytical procedures described herein.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I affirm that I am not engaged in any form of collaboration with other individuals and that all work has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project undertaken toward course fulfilment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.



Candidate Signature

Arun Sankar E K

Date: 13-06-2026

2. Executive Summary

I am working on a project titled "Business Performance Analysis and Service Optimization: A Primary Data Study of a Professional Laundry Business". This project focuses on the business operations of RDS Associates, a privately owned company which is focused on dry cleaning and laundry services. This is a B2C business organization operating since 2022, located in Thrissur district on Kerala.

This Business providing multiple garment services including wash and iron, wash and fold, dry cleaning for various garment types, shoe cleaning, bag cleaning, curtain cleaning, and blanket washing with free pickup and delivery in a limited area.

This shop operates with six staff and works from Monday to Saturday, 09.00 AM to 7.00 PM. Daily revenue generated by the business is approximately 8000 rupees, against monthly costs full rupees 1,90,000. Washing, ironing and drying services are the most popular services provided by the business. The business stores all the data in digitalized format, which helps them to better manage the reporting of customer orders and transactions.

The shop maintains the customer satisfaction rate of over 90 percentage and the complaints are sorted out either directly or by rectifying their services or compensating them for the same.

3. Organization Background

RDS Associates is a privately owned dry-cleaning and laundry service company. The enterprise is an independent B2C business organization operating since 2022, situated at MM Towers, Fatima Nagar, East Fort, Thrissur, Kerala. This Business providing multiple garment services including wash and iron, wash and fold, dry cleaning for various garment types, shoe cleaning, bag cleaning, curtain cleaning, and blanket washing. Additionally, the shop provides doorstep pickup and delivery services to enhance customer convenience. Free pickup and delivery services are offered on a five km radius, while a normal service charge is applied beyond the range.

The shop cooperated with six staff and functioning Monday to Saturday, 09.00 AM to 7.00 PM. The business produces average 8000 rupees daily revenue, against monthly costs full rupees 1,90,000/-. Wash, ironing and drying are the most demanded services in the business. The business is storing all the data in digitalized format, which helps them to maintain the systematic report-keeping of orders and transactions of customers.

The shop maintains the customer satisfaction rate of over 90 percentage and resolves the complaints directly or through service rectification and providing compensation if required.

4. Problem Statement

Even though RDS Associate is a successfully running business, there is a gap in the systematic Analytical understanding about their service demand pattern, contribution to revenue, efficiency in operation, and customer retention behavior, reducing the scope of decision making based on the analysis of data.

4.1 Problem Statement 1: Revenue and Service Demand Imbalance

The business is offering multiple services like wash and iron, wash and fold, dry cleaning, shoe cleaning, bag cleaning, curtain cleaning, and blanket cleaning with various price levels. But there is no systematic analysis of which service is generating more revenue with high contribution, or which is not much utilized. This is making the process of resource prioritization more difficult and inefficient.

4.2 Problem Statement 2: Operational Efficiency and Turnaround Time

Even though there is a four-day TAT, there is a gap of data driven analysis and understanding of how the type of service, volume of orders, and recurring challenges such as labor shortages and power failures affect the performance of delivery of the items.

4.3 Problem Statement 3: Customer Retention and Behavioural Patterns

With only ten regular customers out of an average twenty-five customers daily, also the absence of analysis of purchase behaviour or frequency patterns to increase and promote retention efforts.

5. Background of the Problem

5.1: Revenue and Service Demand Imbalance

The business offering ten various services which includes wash and iron at rupees one hundred and twenty-five to blanket washing which prices varies but lies around four hundred and forty-five. Even though the shop is providing this many services the major demand is concentrated only on wash, iron and dry washing, which creating less demand for the premium and specialized services and those are still underutilized. The absence of a structured analysis of service contribution to the business, the management of the business is unable to determine whether the pricing of the services are optimal or whether the resources, staffs and chemicals are allotted ineffectively.

The business does not currently have a structured analysis of the revenue contribution from each service category. This limits the management's ability to determine whether the pricing structure is optimal, whether certain services are operating at a loss after accounting for labour and material costs, and whether promotional efforts should be directed toward high-margin but underutilized services such as express dry cleaning or specialty item care. Without this understanding, resources such as chemicals, equipment time, and staff allocation may not be directed most efficiently.

5.2 Operational Efficiency and Turnaround Time

Even though there is a four-day TAT, which is around the peak hours of a day between 9am to 11 am and 5pm to 7pm, also peak days on Wednesday and Saturday. The business is facing huge challenge from the labour shortage and power failures which reduce the capacity of process the service and disrupt the operations which depend upon various equipment which include iron and machine washing, increase the risk of delaying and produce inconsistency in delivering.

5.3 Customer Retention and Behavioural Patterns

There will be 25 approximately customers at the shop but there will be approximately only 10 regular customers visiting, indicating a low ratio of retention. The reason behind this low retention rate is not analytically understood. There is a high absence of the customer visiting frequency analysis, preferences of services and the pattern of spending over different groups of customers. These behaviors patterns should be analyses through descriptive analysis and customer segmentation based on visit frequency and usage of services, the shop may not be understand which customers are at the risk of reducing revisit to the shop or affect the targeted methods to improve loyalty and retention.

6. Problem Solving Approach

6.1 Data Collection

The primary data for the study will be directly collected from RDS Associates by using the questionnaires and personal interviews. The data collection will be across two categories, which include Operational_Data and Customer Order Data. The information on the operational data is already available from the business shop through direct interviews and using questionnaires. This includes the average daily revenue, number of daily customers, peak hours of a day and peak days in a week, monthly expenses for the operation, turnaround times of standard and express times, and price list of all services.

In the Customer Order Data, will collect customer order data between 50-100 records, which include the order date, customer identifier, type of customer, which is regular or new, type of service, number of garments, amount in bill, payment mode, delivery data and complaint status. This data set will be the fundamental basis for the statistical analyses.

6.2 Methods

Mainly applying **descriptive analysis** for calculating mean, median, mode, frequency distribution, and percentage analysis will be used to understand the pattern of the demand in different services, the distribution of revenue across services, turnaround time performance, and frequency of visit by customers. This will be a fundamental method of this project to understand the customer behaviour patterns, which include the service preference by customers and spending structure across customer groups, which helps to directly address the customer problems.

To understand the relationship between different variables, will do the **correlation analysis** which includes the number of customers and daily revenue, peak hours and order volume, type of service and turnaround time, and bill amount. This will help to understand which variable will have most effect on the revenue and efficiency of service delivery.

To identify the day and week-wise revenue trends, will use the **Time Series Analysis**. This will help to address the peaks, lows revenue periods, and impact of disruptions in various operations due to the labour shortage and power failures on the performance of the business. Ultimately this analysis will help to the more efficient staff management and planning of resources.

6.3 Tools Used

Microsoft Excel: The Microsoft Excel tool primary used for the organization if collected data, data entry, cleaning, systematic organization. The major formulas of descriptive statistics including AVAREGE, MEDIAN, MODE, COUNTIF and FREQUENCY will be applied. Pivot table will be used for summarising the service wise revenue and customer visit frequency. For the EDA (exploratory data analysis) will use the bar charts, pir charts, and trend lines.

Stata: Stata will be the tool for the doing the statistical analysis. This will be used to produce the descriptive statistics, and for identity the relation between different variables through correlation analysis. Stata will be used for generating Time series plats for revenue trends. Stata will be also used for doing tabulations for service demand analysis and frequency distribution charts to visualise the results more efficiently.

7. Expected Timeline

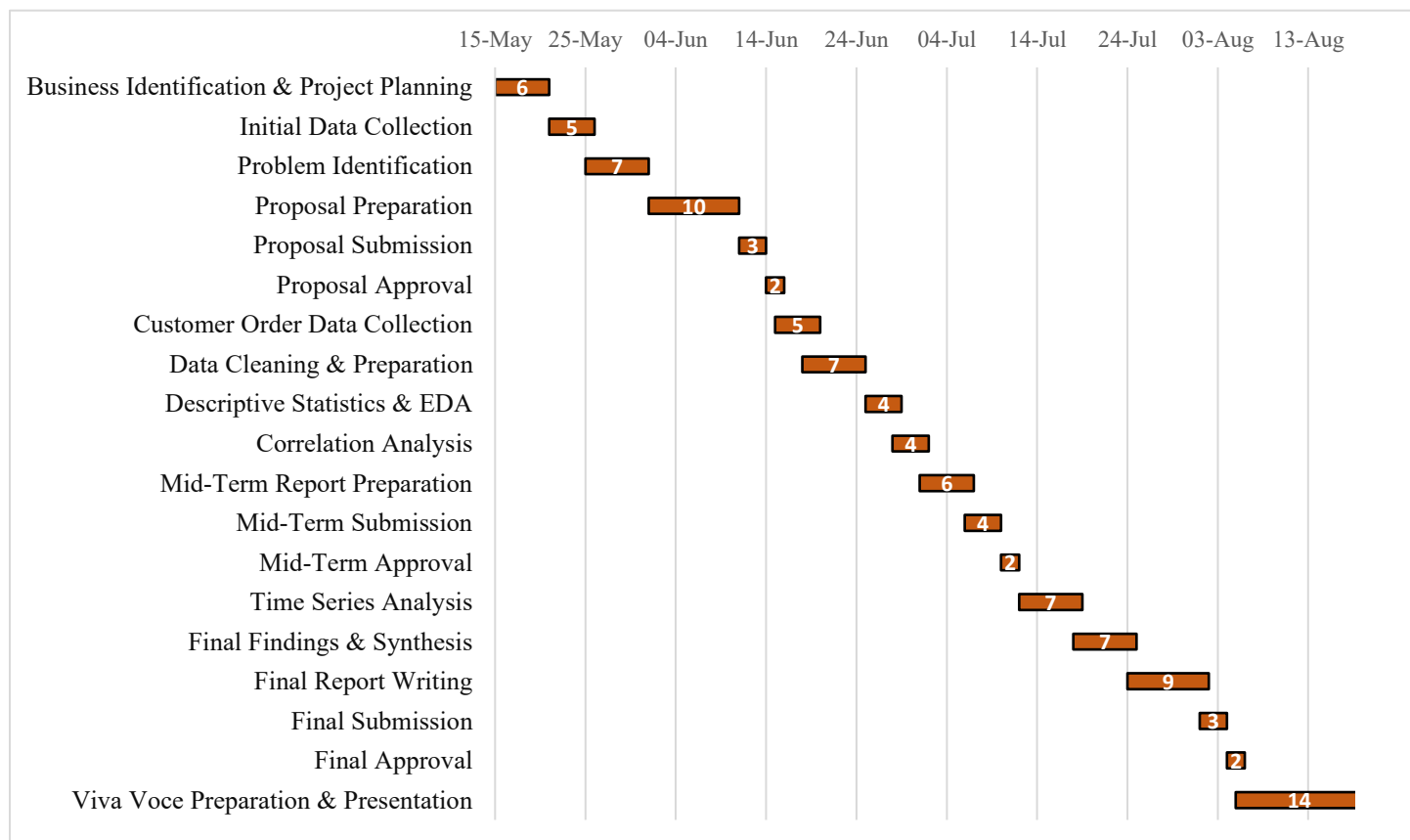
7.1 Work Breakdown Structure (WBS)

The project is organized into four phases: Proposal, Data Collection, Mid-Term, and Final. The Work Breakdown Structure below outlines the key activities under each phase.

Phase	Tasks / Activities
Proposal Phase	• Business identification and Project Planning • Initial data collection • Problem identification • Proposal preparation and writing • Proposal submission
Data Collection Phase	• Collection of customer order data • Collection of daily/weekly revenue data • Service-wise demand and turnaround time data • Operational data (expenses, staff, working hours)
Mid-Term Phase	• Data cleaning and preparation • Descriptive statistics and EDA • Correlation analysis • Progress report and preliminary findings • Mid-term submission
Final Phase	• Time series analysis (weekly revenue trends) • Synthesis of findings and insights • Final report preparation • Final submission • Viva voce presentation

7.2 Gantt Chart

The following Gantt chart shows the expected timeline for the completion of each project activities as well as their submission and approval milestones for all phases.



8. Expected Outcomes

The analysis that carried out from this project is expected to generate the following outcomes, each of which will provide data regarding the business performance of RDS Associates:

8.1 Service Demand and Revenue Distribution Profile

The service wise revenue contribution will be produced from pivot table and descriptive statistics.

This will identify the which services - such as wash and iron, dry cleaning, or shoe cleaning is most share in revenue and which of them are underutilized. The demand patterns will be identified by hr frequency distribution tables across the service portfolio.

8.2 Peak-Period and Operational Efficiency Insights

Correlation analysis between order volume, day of the week, time of day, and turnaround will reveal the relation between service delivery performance and peak period of the business. Time series analysis of weekly revenues will identify patterns related to high revenue and low revenue periods. In this case, it will be easier for management to prepare their schedules of staffs and allocate resources more effectively.

8.3 Customer Segmentation Based on Purchase Behaviour

The performances of services, the frequency of visit, and the pattern between the new customers and regular customers will be obtained by the descriptive analysis. This will help the analyst to understand the behaviour of the customer and understand which group of customers need more engagement and address their issues.

8.4 Data-Driven Foundation for Business Decision-Making

Altogether, the findings from all these analyses will generate evidence-based grasping of the operations. This will help for the most effective decisions regarding the pricing of services in the business, appointment of staffs, working of operations, and the engagement with the customers, which will help the business to move towards more data-based management and systematic working style of the shop.